CATEGORY BASED DAILY EXPENSE MANAGEMENT SYSTEM

Using C++ Programming Language

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1. COVER PAGE

CATEGORY BASED DAILY EXPENSE MANAGEMENT SYSTEM

A Mini Project Report Submitted in Partial Fulfillment
of the Requirement for the Award of Degree

Bachelor of Computer Applications (BCA)

Submitted By:
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Course: BCA
Year: 1st Year
Subject: C++ Programming

College:
Dev Bhoomi Uttarakhand University
Department: School of Engineering & Computing (SOEC)

Academic Session: 2025 – 2026

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2. CERTIFICATE

CERTIFICATE

This is to certify that the project entitled
“Category Based Daily Expense Management System using C++”
submitted by **Surya Prakash**, student of **BCA 1st Year**, **School of Engineering & Computing (SOEC)**, **Dev Bhoomi Uttarakhand University**, is a Bonafide record of work carried out under my guidance during the academic session **2025–26**.

This project is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Computer Applications (BCA)**.

Project Guide:

Miss. Dhruvi Dagar

Department: SOEC

Date: 27th April 2026
Place: Dehradun

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3. ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Dev Bhoomi Uttarakhand University** for providing a supportive academic environment.

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Lastly, I acknowledge the resources such as books, faculty notes, and online learning platforms that helped me successfully complete this project.

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4. TABLE OF CONTENTS

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5. ABSTRACT

The **Category Based Daily Expense Management System** is a console-based C++ application designed to help users record, manage, and analyze their daily expenses based on different categories such as food, travel, shopping, and bills.

This project focuses on simplifying expense management for students and individuals by offering an easy-to-use interface that stores expenses during program execution and calculates category-wise totals.

The project also helps a beginner to understand fundamental concepts of **C++ programming** such as structures, arrays, functions, loops, and conditional statements.

6. INTRODUCTION

6.1 Expense Management

Expense management refers to the process of recording, tracking, and controlling daily expenditures. Without proper tracking, individuals may overspend unknowingly.

6.2 Need for Expense Tracking

Manual tracking using notebooks is time-consuming and prone to errors. A computerized system ensures accuracy and quick calculations.

6.3 Importance for Students

Students usually have a fixed budget. Tracking expenses helps them:

- Avoid unnecessary spending
- Save money
- Understand spending habits

6.4 Why C++ is Used

C++ is chosen for this project because:

- It is easy to learn for beginners
- Supports structured programming
- Widely used in academic curriculum
- Efficient and fast

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7. PROBLEM STATEMENT

Most people forget their daily expenses and cannot analyze category-wise spending using manual methods. This leads to poor financial planning and uncontrolled expenses.

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8. OBJECTIVES OF PROJECT

- To record daily expenses
- To categorize expenses
- To calculate total expenditure
- To provide menu-based interaction
- To learn practical C++ programming

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9. SCOPE OF PROJECT

- Personal daily expense tracking
- Student budget planning
- Small household usage
- Academic mini project
- Learning console-based application development

10. SOFTWARE / HARDWARE REQUIREMENTS

Software Requirements

Software	Description
OS	Windows
Compiler	Turbo C++ / CodeBlocks / VS Code
Language	C++

Hardware Requirements

Hardware	Minimum Requirement
Computer	Basic Laptop / Desktop
Input Devices	Keyboard, Mouse

11. CONCEPTS USED IN C++

11.1 Variables

Used to store data like amount, choice, category.

11.2 Structure

Used to group expense details like category and amount.

11.3 Arrays

Used to store multiple expense records.

11.4 Functions

Used to organize code into modules.

11.5 Loops

Used to repeat menu and calculations.

11.6 If-Else Statements

Used for decision making.

11.7 Input / Output

Used to take input and display output.

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12. SYSTEM DESIGN

12.1 Flowchart

Start

|

Display Menu

|

User Choice

|

Add / View / Total

|

Repeat

|

Exit

12.2 Algorithm

1. Start the program
2. Display menu options

3. Accept user choice
4. Perform selected operation
5. Repeat until user exits

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13. MODULES OF PROJECT

Add Expense Module

Stores category and amount.

View Expense Module

Displays all recorded expenses.

Category Total Module

Calculates total for:

- Food
- Travel
- Shopping
- Bills

Exit Module

Safely terminates the program.

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14. SOURCE CODE

C++

```
#include <iostream>

using namespace std;


struct Expense {

int category;

float amount;

};
```

```
Expense e[100];
int count = 0;

void addExpense() {
cout << "Select Category:\n";
cout << "1. Food\n2. Travel\n3. Shopping\n4. Bills\n";
cin >> e[count].category;

cout << "Enter Amount: ";
cin >> e[count].amount;

count++;
cout << "Expense Added Successfully!\n";
}

void viewExpense() {
cout << "\nAll Expenses:\n";
for (int i = 0; i < count; i++) {
cout << "Category: " << e[i].category
<< " Amount: " << e[i].amount << endl;
}
}

void categoryTotal() {
float total[4] = {0};

for (int i = 0; i < count; i++) {
total[e[i].category - 1] += e[i].amount;
}
```

```
cout << "\nCategory Totals:\n";
cout << "Food: " << total[0] << endl;
cout << "Travel: " << total[1] << endl;
cout << "Shopping: " << total[2] << endl;
cout << "Bills: " << total[3] << endl;
}

int main() {
int choice;
do {
cout << "\n1. Add Expense\n2. View Expense\n3. Category Total\n4. Exit\n";
cout << "Enter Choice: ";
cin >> choice;

switch (choice) {
case 1: addExpense(); break;
case 2: viewExpense(); break;
case 3: categoryTotal(); break;
case 4: cout << "Thank You!\n"; break;
default: cout << "Invalid Choice!\n";
}
} while (choice != 4);

return 0;
}
```

Show more lines

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15. OUTPUT SCREENS

- Main Menu Screen
- Add Expense Screen
- View Expense Screen
- Category Total Screen

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16. ADVANTAGES

- User-friendly
- Fast and accurate
- Organized records
- Beginner-friendly code
- Educational project

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17. LIMITATIONS

- No permanent storage
- Console-based interface
- No authentication
- Limited categories

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18. FUTURE ENHANCEMENTS

- File handling support
- Login system
- Monthly reports
- Graphical interface

- Mobile application
- Charts and analytics

19. CONCLUSION

The **Category Based Daily Expense Management System** is a successful mini project that fulfills its objectives effectively. It provides hands-on experience in C++ programming and helps understand real-world problem solving using structured programming concepts.

20. REFERENCES

1. Bjarne Stroustrup – *The C++ Programming Language*
2. E. Balagurusamy – *Object Oriented Programming with C++*
3. Faculty Notes – SOEC
4. Online Tutorials (GeeksforGeeks, TutorialsPoint)